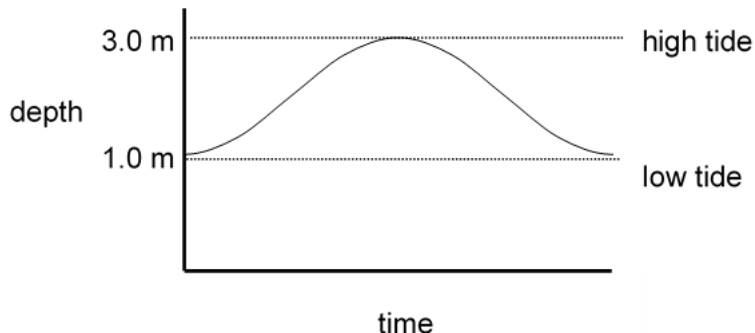


- 15** The rise and fall of water in a harbour is simple harmonic. The depth varies between 1.0 m at low tide and 3.0 m at high tide. The time between successive low tides is 12 hours.



A boat, which requires a minimum depth of water of 1.5 m, approaches the harbour at low tide. How long will the boat have to wait before entering?

- | | |
|----------|-----------|
| A | 0.5 hours |
| B | 1.0 hours |
| C | 2.0 hours |
| D | 2.5 hours |

Ans: C

$$-0.5 = \cos(2\pi/12)t \text{ and thus } t = 2.0 \text{ hrs}$$

- 16** When coherent monochromatic light falls on double slits, an interference pattern is observed.